A Training-Free Dynamic Surface Distance Field for Efficient Finite-Element Contact of Deformable Bodies

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Abstract

Signed distance fields provide an attractive geometric representation for contact detection because distance, penetration depth, and contact normals can be obtained from local field queries. However, classical SDFs are typically static and therefore difficult to apply directly to deformable finite-element bodies whose contact surfaces evolve with the nodal degrees of freedom. Existing dynamic SDF approaches often rely on precomputed deformation snapshots, reduced bases, or neural approximations, which introduce training costs and generalization concerns.

This paper presents a training-free dynamic surface-distance framework for finite-element contact. Instead of reconstructing a deformed SDF from data, the proposed method induces a local oriented surface distance directly from the current finite-element boundary. The contact gap is defined as the signed distance from a slave surface sample to the current master surface, and the corresponding normal, closest-point projection, contact Jacobian, and penalty contact force are assembled consistently from the current FEM geometry. A standalone implementation is developed without dependence on Abaqus-generated matrices, force files, or post-processing outputs.

The implementation is validated primarily on C3D4/TET4 and C3D8 finite elements with normal penalty contact, including linear and clean-room StVK/HHT validation backends. Verification is organized in three layers: analytic and scikit-fem checks for the non-contact FEM stress and displacement fields, CalculiX contact-output comparisons for static C3D8 contact energy and dynamic C3D4/C3D8 contact trajectories, and implementation-level closest-point checks for the accelerated distance query. In the final non-quick experiments, the dynamic FEM-induced surface-distance pipeline achieves speedups from 3.58× to 57.07× over brute-force all-triangle projection for the tested surface resolutions; a separate total-step timing check also remains faster than the brute-force counterpart in the tested cases. The current scope is limited to C3D4/TET4 and C3D8 validation backends, oriented local surface distances, spatial-hash broad phase, and frictionless penalty normal contact.

1 Introduction

Contact handling is a central component of deformable finite-element simulation. At each time step, a solver must identify potential contact pairs, evaluate separation or penetration, compute contact normals, and assemble force and stiffness contributions into the global system. These operations are geometric as well as mechanical: an inaccurate gap or normal can lead to incorrect contact activation, force direction errors, or unstable time integration. As surface resolution increases, the repeated closest-point search between deformable surfaces can become a significant computational bottleneck.

Signed distance fields offer an attractive representation for contact because a local query can provide distance, penetration depth, and normal direction. However, a conventional SDF is usually defined for a fixed shape. This assumption is not compatible with finite-element bodies whose boundary surfaces move with nodal degrees of freedom. A distance field constructed in the reference configuration generally does not remain a true Euclidean distance field after stretch, compression, or shear. Therefore, directly reusing a material-space SDF can introduce errors in both the gap value and the contact normal.

Dynamic or deformable SDF representations can address this issue by approximating how the distance field changes with deformation. Many such approaches rely on precomputed deformation examples, reduced-order bases, learned neural fields, or other data-driven approximations [2, 6]. These methods can be effective when the deformation space is well sampled, but they introduce offline preparation costs and may generalize poorly to deformation modes outside the training or basis space. For mechanics-oriented simulation, this creates a gap between the finite-element state, which is available at every step, and the distance representation, which may be only indirectly tied to the current surface geometry.

This paper proposes a training-free dynamic surface-distance framework for deformable FEM contact. Instead of reconstructing a deformed volumetric SDF from data, the method induces an oriented local distance directly from the current finite-element boundary surface. A spatial-hash broad phase first identifies candidate boundary triangles, and a local closest-point projection then evaluates the gap, closest point, barycentric weights, and surface normal on the current deformed surface. The resulting distance query is therefore tied directly to the current FEM configuration.

The current-surface distance is used consistently in the contact formulation. The contact gap is defined as the signed distance from a slave surface sample to the current master surface. The slave and master Jacobian blocks are assembled from the same normal and shape-function weights used in the closest-point projection. A frictionless penalty contact force and Gauss-Newton contact stiffness approximation are then assembled from this Jacobian. This gives a compact contact pipeline in which geometry evaluation and force assembly share the same local surface representation.

We implement the method in a standalone FEM-SDF prototype with internally assembled element stiffness, mass, body-force, constraint, contact, and time-integration components. The core solver does not depend on Abaqus-generated matrices, force files, `.odb` files, or spreadsheet outputs. Experiments first verify the non-contact FEM backend against analytic stress/strain references and an equivalent scikit-fem model. The contact evidence then uses external CalculiX outputs: a static C3D8 contact-energy test validates force and energy replay on the deformed current surface using the registered C3D8 stiffness backend; a dynamic C3D4 block-plane case compares activation time, penetration, normal reaction, contact energy, center-of-mass motion, and contact-zone stress; and generated C3D8 block-plane/block-block trajectories test whether the same dynamic SDF replay remains accurate on C3D8 current surfaces. Brute-force closest-point projection is retained only as an implementation-level geometry oracle and timing baseline. In the final non-quick experiments, the dynamic FEM-induced surface-distance pipeline achieves speedups from $3.58\times$ to $57.07\times$ over brute-force all-triangle projection for the tested surface resolutions.

In summary, this work makes four contributions. First, it introduces a training-free current-surface-induced distance representation for deformable FEM contact. Second, it derives a consistent contact gap, normal, Jacobian, and penalty force assembly from local current-surface projection. Third, it provides a standalone FEM-SDF implementation with registered C3D4/TET4 and C3D8 mechanics backends independent of Abaqus-generated solver artifacts. Fourth, it presents a reproducible validation and evidence package in which FEM field correctness is checked against analytic and scikit-fem references, contact correctness is checked against CalculiX contact outputs on C3D4 and C3D8 surfaces, and computational benefit is measured against brute-force all-triangle projection.

2 Related Work

Finite-element contact methods typically rely on geometric proximity queries between deformable surfaces, followed by the assembly of contact constraints or penalty forces [3]. Classical node-to-surface and mortar formulations compute gaps and normals from closest-point projections on the current or predicted contact surface [5]. These approaches are mechanically direct, but repeated all-pair or all-triangle closest-point projection becomes expensive as the number of candidate surface features grows. Our method keeps the current-surface projection principle, but combines it with a dynamic surface-distance query and spatial hashing to avoid brute-force all-triangle projection in the tested pipeline.

Signed distance fields have been widely used for collision detection and contact response because they provide a convenient way to query distance and normal information [7]. For rigid or fixed geometries, an SDF can be precomputed and queried efficiently during simulation. The difficulty arises when the body is deformable: the contact surface changes with the finite-element state, while a static reference SDF does not generally preserve current-configuration Euclidean distances. Our material-space baseline experiments show this issue directly under stretch and shear deformation, where the carried reference-space distance produces nonzero gap and normal errors.

Several dynamic SDF approaches address deformation by learning or approximating the evolution of the distance field from examples, reduced coordinates, or neural representations [2, 6]. These methods can provide fast queries after training, but their accuracy depends on the deformation space represented by the data or model. In contrast, the method proposed here does not train a distance model and does not require deformation snapshots. The distance is induced at query time from the current FEM boundary, which makes the geometric representation directly consistent with the nodal configuration used by the mechanics solver.

Efficient contact pipelines usually separate broad-phase candidate detection from narrow-phase geometric evaluation. Common broad-phase structures include uniform grids, spatial hashes, bounding-volume hierarchies, and related acceleration structures [1]. The current implementation uses a uniform spatial hash over padded triangle AABBs. This choice is simple and deterministic, and it is sufficient to demonstrate the paper's core comparison against brute-force all-triangle projection. We do not claim superiority over optimized production BVH, IPC collision pipelines, GPU collision detection, or commercial contact implementations [4].

The proposed method sits between traditional closest-point contact and precomputed or learned SDF contact. Like closest-point FEM contact, it evaluates gap and normal from the current deformed surface. Like SDF-based contact, it presents the narrow-phase query as a local distance evaluation. Unlike static or data-driven SDF approaches, it does not rely on a reference field, deformation training data, neural approximation, or external Abaqus-generated matrices. The result is a training-free and reproducible FEM-SDF contact prototype for the specific setting of linear TET4 elasticity and frictionless penalty normal contact.

3 Method

3.1 Current finite-element surface

Let the deformable body occupy a reference domain Ω_0, discretized by finite elements. The current nodal coordinates are denoted by $\mathbf{x}_a$. The contact query acts on a triangulated boundary surface: a TET4 boundary face is already triangular, while a C3D8 boundary quad is split into triangles for the SDF query. A boundary triangle e on the current surface is represented as

$$\mathbf{s}_e(\boldsymbol{\xi}, \mathbf{q}) = \sum_{a \in I_e} N_a^\Gamma(\boldsymbol{\xi}) \mathbf{x}_a, \tag{1}$$

where $\boldsymbol{\xi}$ denotes surface coordinates, N_a^Γ are the surface shape functions, and $\mathbf{q}$ collects the nodal degrees of freedom. The outward unit normal on the current surface patch is

$$\mathbf{n}_e = \frac{\mathbf{s}_{,\xi} \times \mathbf{s}_{,\eta}}{\|\mathbf{s}_{,\xi} \times \mathbf{s}_{,\eta}\|}. \tag{2}$$

The implementation repairs negatively oriented TET4 elements and orients boundary faces outward for closed TET4 surfaces, ensuring a consistent local sign convention for the supported C3D4 trajectory experiments. C3D8 contact replay uses the same current-surface triangle query after boundary quads are triangulated.

3.2 Dynamic surface distance

For a slave query point $\mathbf{x}_i^A$, the master body B defines a current surface-induced distance

$$g_i = \phi_B(\mathbf{x}_i^A, \mathbf{q}_B). \tag{3}$$

In the implementation, ϕ_B is evaluated by first obtaining candidate boundary triangles from a spatial hash broad phase and then computing the closest point on the candidate current-surface triangles. If $\mathbf{p}_i^*$ is the closest point and $\mathbf{n}_i$ is the corresponding oriented surface normal, the local signed gap is evaluated as

$$g_i = (\mathbf{x}_i^A - \mathbf{p}_i^*) \cdot \mathbf{n}_i. \tag{4}$$

A positive gap denotes separation, while a negative gap denotes penetration. This formulation should be interpreted as an oriented local surface distance induced by the current FEM surface. It is not claimed to be a robust global signed distance method for arbitrary open, non-manifold, or inconsistently oriented geometries.

3.3 Contact Jacobian

Let the slave query point be interpolated from slave body nodes as

$$\mathbf{x}_i^A = \sum_{b \in I_A} N_b^A \mathbf{x}_b^A. \tag{5}$$

Let the closest master point be

$$\mathbf{p}_i^* = \sum_{a \in I_B} N_a^B \mathbf{x}_a^B. \tag{6}$$

The first variation of the gap is

$$\delta g_i = \mathbf{n}_i^T \delta \mathbf{x}_i^A - \mathbf{n}_i^T \delta \mathbf{p}_i^*. \tag{7}$$

Therefore,

$$\frac{\partial g_i}{\partial \mathbf{x}_b^A} = N_b^A \mathbf{n}_i^T, \tag{8}$$

and

$$\frac{\partial g_i}{\partial \mathbf{x}_a^B} = -N_a^B \mathbf{n}_i^T. \tag{9}$$

These expressions define the contact Jacobian row $\mathbf{J}_i$.

3.4 Penalty contact force

For frictionless normal contact, a penalty energy is used:

$$E_{c,i} = \frac{1}{2} k_i \langle -g_i \rangle_+^2, \tag{10}$$

where

$$\langle x \rangle_+ = \max(x, 0). \tag{11}$$

The normal contact multiplier is

$$\lambda_i = k_i \langle -g_i \rangle_+. \tag{12}$$

The global contact force contribution is then

$$\mathbf{f}_{c,i} = \mathbf{J}_i^T \lambda_i. \tag{13}$$

A Gauss-Newton approximation to the contact stiffness may be assembled as

$$\mathbf{K}_{c,i} \approx k_i \mathbf{J}_i^T \mathbf{J}_i. \tag{14}$$

4 Implementation

The method is implemented as a standalone Python prototype. The solver contains modules for mesh topology, registered element backends, analysis backends, boundary extraction, current-surface distance queries, contact broad phase, contact narrow phase, penalty force assembly, and validation scripts.

The finite-element implementation supports small-strain linear C3D4/TET4 and HEX8/C3D8 elasticity through a common backend interface. For TET4, the element stiffness is assembled as

$$\mathbf{K}_e = V_e \mathbf{B}_e^T \mathbf{C} \mathbf{B}_e, \tag{15}$$

where V_e is the TET4 volume, $\mathbf{B}_e$ is the strain-displacement matrix, and $\mathbf{C}$ is the isotropic linear-elastic constitutive matrix. The C3D8 backend uses the corresponding trilinear shape functions, engineering-strain $\mathbf{B}$ matrix, and $2 \times 2 \times 2$ Gauss integration used by the external C3D8 validation model. Both consistent and lumped mass options are implemented, and gravity/body-force assembly is performed internally.

The broad phase uses a uniform spatial hash over padded triangle AABBs. Candidate triangles are passed to the dynamic surface-distance query. A brute-force all-triangle closest-point projection is retained only as an implementation-level geometry oracle and timing baseline; external contact-solver outputs provide the main contact-correctness evidence.

The implementation is independent of Abaqus in the core solver. No Abaqus `.inp` parser, Abaqus-generated global stiffness/mass matrices, `.mat` force files, `.odb` files, or spreadsheet outputs are required by the core `src/sfc` package. The final Phase-6 experiment package confirms that no `src/sfc` files were changed during final result packaging.

5 Experiments and Results

The experiments are organized to separate mechanics correctness, contact correctness, and acceleration evidence. This separation is important because the proposed contribution is not a replacement

Table 1: Validation roadmap. External solver outputs provide the primary contact-correctness evidence; brute-force closest-point projection is restricted to implementation checks and timing baselines.

Claim family	Evidence source	Role in the paper
FEM stress/displacement	Analytic patch and scikit-fem	Non-contact mechanics correctness
Static contact law and energy	CalculiX `contactenergy.inp`	External contact-output reference
Official contact-law alignment	CalculiX `contact1/contact3/contact6`	Deformed-state SDF gap and pressure-law replay
Official engineering candidates	CalculiX `scheibe2f2f` and `ball`	Future native SFC comparison targets, not current claims
Dynamic contact trajectory	CalculiX generated C3D4 block-plane case	External trajectory reference
C3D8 backend extensibility	SFC companion solve on `contactenergy.inp`	Linear C3D8 backend evidence
Dynamic C3D8 surface replay	CalculiX generated C3D8 block-plane/block-block cases	External trajectory replay on C3D8 current surfaces
Current-surface SDF geometry	CalculiX-deformed geometry replay	Gap/force/energy on deformed surface
Query acceleration	Brute-force all-triangle projection	Timing baseline only

for FEM mechanics; it is a current-surface distance query used inside a contact formulation. Therefore, the non-contact displacement and stress fields are first checked against analytic and external FEM references. Contact correctness is then evaluated against external solver contact outputs, primarily CalculiX reaction, contact energy, penetration, active contact count, and stress diagnostics. Brute-force closest-point projection is used only as an implementation-level geometry oracle and as the performance baseline for contact search cost.

The final paper-scale performance and material-space baseline results are taken from the Phase-6 non-quick experiment package under `results/paper_final`. Physical-comparison evidence is generated by the Phase-7 package and the locked paper numerical experiments under `paper/numerical_experiments`. We also maintain a conservative official CalculiX example line: `contactenergy.inp` is the direct C3D8 static contact reference; `contact1/contact3/contact6` now provide official deformed-state SDF gap and pressure-overclosure replay diagnostics against CalculiX `CDIS/CSTR`; and `scheibe2f2f.inp.gz` plus `ball.inp.gz` are recorded as future engineering candidates until native SFC metric-level comparisons are implemented. The recorded Phase-6 runtime metadata is Python 3.13.5, NumPy 2.1.3, SciPy 1.15.3, Matplotlib 3.10.0, Windows 11 AMD64, and 24 logical CPUs.

Because the desired validation space includes static and dynamic contact, linear and geometric-nonlinear response, and multiple element families, the paper uses an explicit claim-gated evidence matrix. Table 2 records the current C3D8 status. A row is allowed to support external correctness only when a native SFC result and a CalculiX comparison both exist. A row is allowed to support trajectory equivalence only when a native SFC trajectory is compared against a CalculiX trajectory. A row is allowed to support efficiency only when timing evidence exists for that same case. This rule prevents replay-only rows from being promoted into broader solver-equivalence claims.

The warped/non-planar row is deliberately kept as a diagnostic row. It runs a CalculiX C3D8 case with a non-planar master surface and replays the exported deformed states with the SFC

Table 2: Phase-9 C3D8 claim-gated evidence matrix. C3D10 is excluded because no C3D10 backend is registered.

Case	Analysis	Response	Native SFC	CalculiX	External correctness	Trajectory equivalence
C3D8 linear static contact	Static	Linear	yes	yes	yes	no
C3D8 nonlinear static contact	Static	Geometric nonlinear	yes	yes	yes	no
C3D8 linear dynamic block-plane	Dynamic	Linear	yes	yes	yes	yes
C3D8 linear dynamic block-block	Dynamic	Linear	yes	yes	yes	yes
C3D8 nonlinear dynamic block-plane	Dynamic	Geometric nonlinear	yes	yes	yes	yes
C3D8 nonlinear dynamic block-block	Dynamic	Geometric nonlinear	yes	yes	yes	yes
C3D8 warped/non-planar replay	Dynamic replay	Geometric nonlinear	yes	yes	no	no

Table 3: Analytic stress/strain patch verification for affine small-strain linear elasticity.

Resolution	Elements	Max strain error	Max stress error
1	5	0.000000×10^0	0.000000×10^0
2	40	0.000000×10^0	0.000000×10^0
3	135	0.000000×10^0	0.000000×10^0
4	320	0.000000×10^0	0.000000×10^0

current-surface query, but the current RF/CELS/CDIS metrics do not pass the external-correctness gate. We therefore use it as evidence that the non-planar comparison path exists, not as a supported trajectory-equivalence or accuracy claim.

5.1 FEM field verification before contact

We first verify the FEM field quantities before evaluating any contact claim. For an affine uniaxial patch, the TET4 strain energy, engineering strain, and stress are compared against analytic small-strain linear elasticity. The maximum reported relative energy error in the Phase-6 package is

$$3.47 \times 10^{-16}. \tag{16}$$

For $E = 2.0 \times 10^5$, $\nu = 0.25$, and affine displacement $u_x = 10^{-3}x$, $u_y = u_z = 0$, the analytic Voigt strain is $[10^{-3}, 0, 0, 0, 0, 0]$, and the analytic stress is $[240, 80, 80, 0, 0, 0]$. Table 3 shows that the recovered strain and stress match the analytic values to reported precision.

As a second mechanics check, a fixed-base cantilever is solved over mesh refinement and against scikit-fem on the same TET4 mesh. The internal refinement trend in Table 4 is included to show expected displacement/stress stabilization, while the external scikit-fem comparison in Table 5 checks the assembled stiffness, displacement, strain, stress, and von Mises fields against an independent implementation. Stress-cloud figures are used only for visual audit; quantitative comparisons use the element stress values in the CSV data.

The maximum von Mises relative error in this comparison is 7.996486×10^{-13}. Figure 5 compares the SFC and scikit-fem stress clouds for the finest tested mesh and shows the absolute stress error. These results verify that the linear TET4 stiffness assembly and resulting stress field agree with an external open-source FEM implementation for the tested equivalent model.

5.2 External static contact reference: CalculiX C3D8 contact energy

The first contact-correctness experiment uses CalculiX's official `contactenergy.inp` surface-to-surface contact test. CalculiX performs the static C3D8 contact solve and prints contact force and contact spring energy quantities. SFC then replays the final CalculiX-deformed geometry with

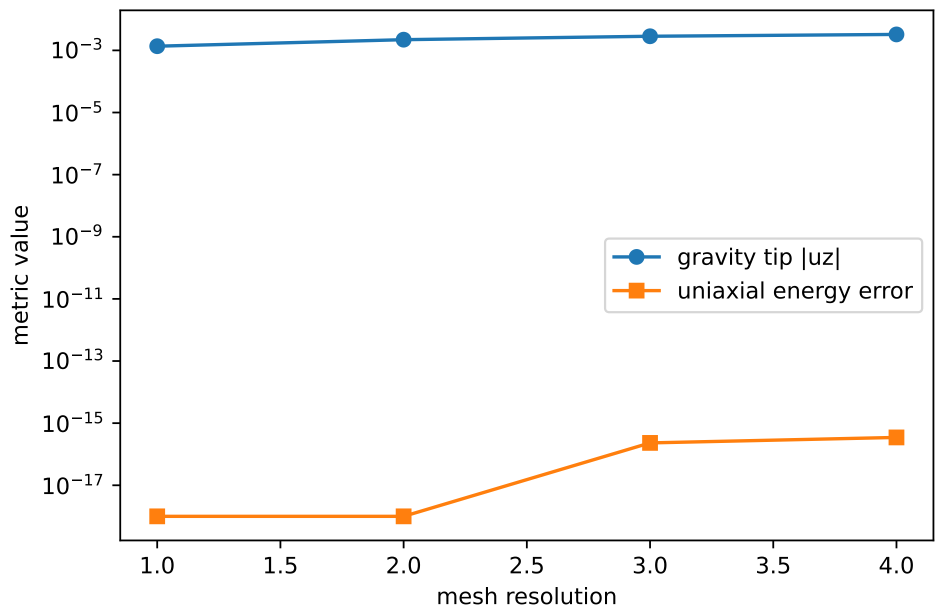

Figure 1: Mesh-resolution trend evidence generated by the Phase-6 package. The plot reports the linear-elastic patch energy error and the gravity-block displacement trend for the tested resolutions.

Table 4: Fixed-base cantilever displacement and von Mises stress trend. Relative errors are measured against the finest internal TET4 run.

Resolution	Elements	Tip u_z	Max von Mises	Tip rel. error
1	5	-1.397105×10^{-3}	9.796707×10^{1}	8.538791×10^{-1}
2	40	-3.686769×10^{-3}	1.835477×10^{2}	6.144070×10^{-1}
3	135	-6.617482×10^{-3}	2.557485×10^{2}	3.078887×10^{-1}
4	320	-9.561298×10^{-3}	3.144287×10^{2}	0.000000×10^{0}

the proposed current-surface dynamic SDF on triangulated C3D8 boundary faces. A companion SFC static solve uses the registered C3D8 stiffness backend with the same linear elastic material, nodal constraints, loads, and pressure-overclosure stiffness. This is an external contact-output comparison: the reference values are CalculiX force and contact-energy outputs, while brute-force projection is not used as the correctness target.

The replay reproduces the CalculiX normal force and contact spring energy to approximately 4×10^{-8} relative error. Because this case uses C3D8 elements, it also checks that the current-surface distance query is not topologically restricted to TET4 boundaries. The C3D8 companion solve is linear static evidence for the registered C3D8 backend; native C3D8 dynamic trajectory evidence is reported in the Phase-9 matrix below.

We further instrument the official CalculiX `contact1`, `contact3`, and `contact6` examples with contact-output requests without changing their physical model. This produces `CDIS`, `CSTR`, and `CELS` rows for small C3D8 node-to-surface contact law checks. Table 7 reports the resulting deformed-state replay errors. These rows validate the current-surface gap and pressure law against CalculiX contact output; they are not native SFC trajectory-equivalence claims.

For a direct field comparison on the same C3D8 model, we additionally assemble an SFC

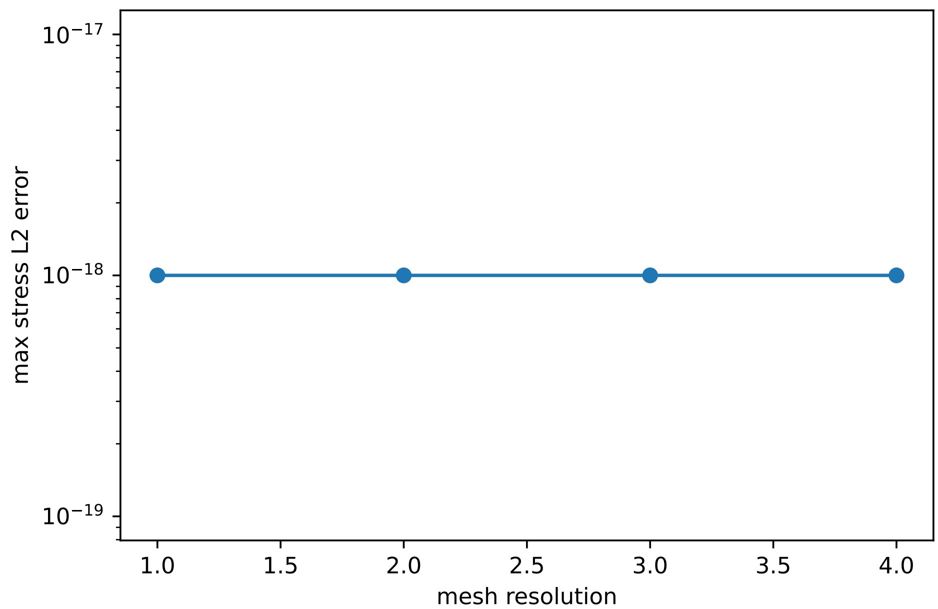

Figure 2: Stress recovery error for the analytic affine strain patch. The error floor reflects exact recovery of the affine field in the tested TET4 meshes.

Table 5: External open-source FEM comparison against scikit-fem 12.0.1 for the equivalent linear TET4 cantilever model.

Resolution	Elements	Disp. rel. error	Stress rel. error	K rel. error
1	5	1.733362×10^{-14}	1.730092×10^{-14}	3.626491×10^{-16}
2	40	5.060386×10^{-14}	5.730537×10^{-14}	5.197130×10^{-16}
3	135	1.095487×10^{-13}	1.276845×10^{-13}	2.713819×10^{-16}
4	320	8.203661×10^{-13}	8.507136×10^{-13}	6.519882×10^{-16}

C3D8 static solve through the registered element backend with the same linear elastic material, constraints, nodal loads, and linear pressure-overclosure stiffness. This allows the SFC C3D8 von Mises field to be plotted against the CalculiX field and exposes the remaining error from the simplified validation contact discretization.

5.3 External dynamic C3D8 contact trajectory replay

To test whether the same current-surface distance replay works beyond triangular C3D4/TET4 surfaces, we additionally generate two transient C3D8 CalculiX contact trajectories. The first case is a C3D8 elastic block contacting a fixed rigid plane; the second is a C3D8 slave block contacting a fixed C3D8 master block. CalculiX supplies the dynamic displacement trajectory, fixed-support reaction force, contact displacement `CDIS`, contact spring energy `CELS`, active contact count, and element stress output. SFC then replays each exported current configuration by triangulating the C3D8 boundary quads and evaluating the dynamic surface SDF on the current master surface.

The replay matches CalculiX `CDIS` with relative errors 1.03×10^{-4} for block-plane and 3.48×10^{-6} for block-block in the locked quick evidence. The force and energy errors in Table 8 are also below

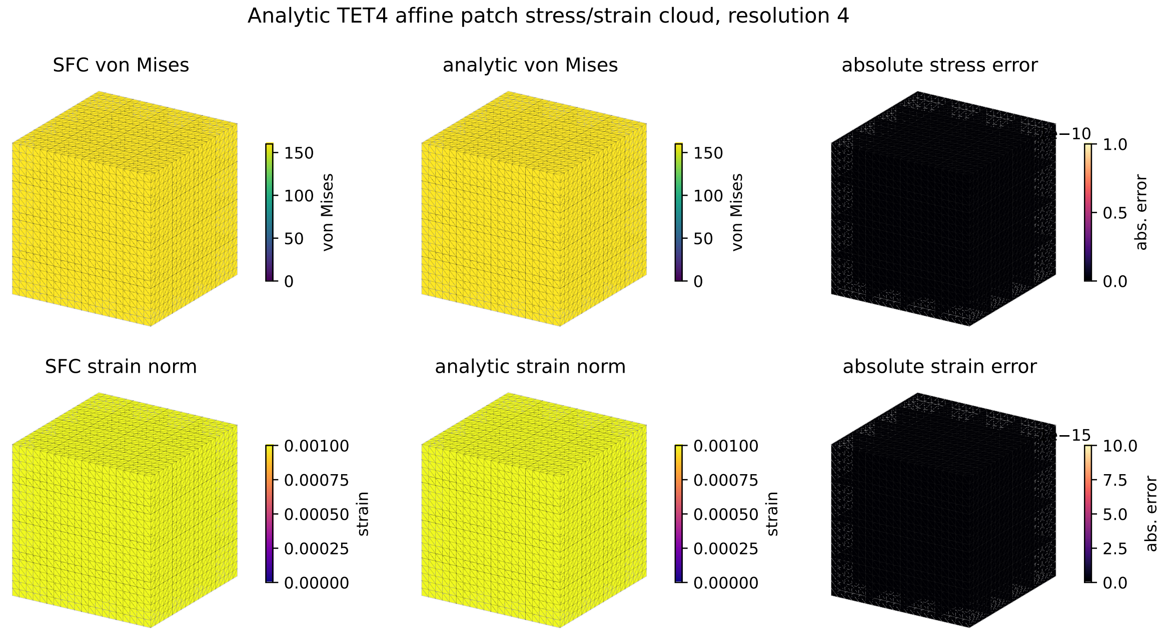

Figure 3: Analytic affine TET4 patch stress/strain cloud on the finest tested resolution. The first row compares SFC and analytic von Mises stress and shows the absolute stress error; the second row compares engineering strain norm and its absolute error. The uniform fields and near-zero error panels are expected for the affine patch.

Table 6: Static external contact-output comparison on CalculiX `contactenergy.inp`. SFC replays the final CalculiX-deformed C3D8 boundary with the current-surface dynamic SDF.

Quantity	CalculiX	SFC dynamic-SDF replay	Relative error
Normal force magnitude	4.000000	3.999999848	3.797933×10^{-8}
Contact spring energy	2.000010×10^{-5}	$2.000009924 \times 10^{-5}$	3.798346×10^{-8}
Maximum penetration	–	1.000005×10^{-5}	–
Active quadrature rows	–	2	–

3×10^{-4}. This supports the geometry and contact-output replay claim for current C3D8 surfaces. The separate native C3D8 dynamic comparison below uses SFC-computed trajectories rather than replaying CalculiX displacements.

5.4 Native C3D8 dynamic trajectory comparison

The Phase-9 evidence package also runs native SFC C3D8 dynamic contact trajectories and compares them against generated CalculiX references in the same block-plane and block-block settings. The linear rows use the C3D8 small-strain stiffness and Newmark average-acceleration path. The geometric-nonlinear rows use the clean-room StVK C3D8 backend with full $2 \times 2 \times 2$ Gauss integration and HHT/Newmark Newton stepping. Contact gap, normal, closest point, Jacobian, and penalty force are still evaluated on the current deformed surface.

The same Phase-9 package includes a C3D4/C3D8 side-by-side CSV table under `paper/numerical_experiments/phase9_full_contact_validation`. This table keeps the C3D4 locked 1 s

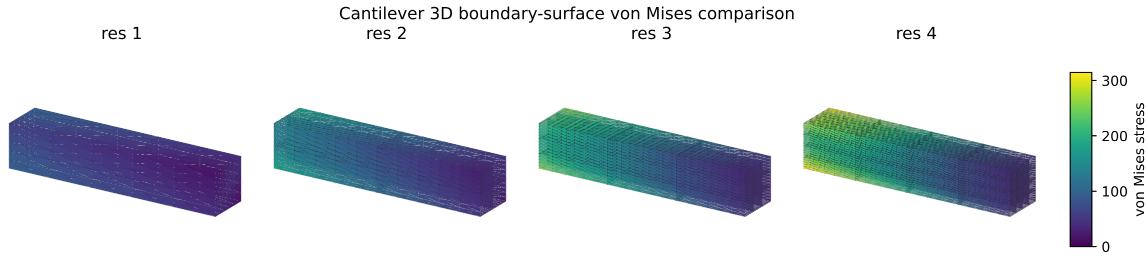

Figure 4: Standard FEM cantilever 3D boundary-surface stress comparison. Element von Mises stresses are averaged to nodes and interpolated over subdivided boundary triangles for visualization; the shared color scale is used across mesh resolutions.

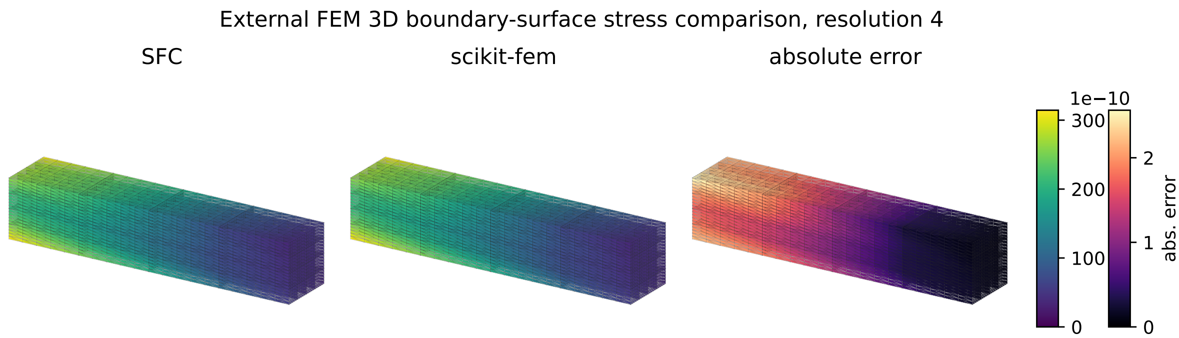

Figure 5: External FEM 3D boundary-surface stress comparison against scikit-fem on the finest tested equivalent cantilever model. The first two panels show nodal-averaged interpolated SFC and scikit-fem von Mises stress on the boundary surface; the third panel shows the interpolated absolute error.

block-plane trajectory and the C3D8 native block-plane/block-block rows in a single schema. We do not merge these rows into a single convergence claim because the element types, meshes, and CalculiX output definitions differ.

Finally, a warped C3D8 master-surface diagnostic is included to exercise a curved/non-planar contact path. It generates a CalculiX trajectory and replays the deformed non-planar C3D8 surface with SFC dynamic SDF. The current RF, CELS, and CDIS metrics do not pass the external-correctness gate, so this row is reported as diagnostic evidence rather than a supported paper claim.

5.5 External dynamic contact trajectory: CalculiX C3D4 block-plane

The next contact-correctness experiment compares a dynamic block-plane contact trajectory against CalculiX. The model uses a generated C3D4 block with matching material, time step $dt = 0.001$, duration 1 s, direct dynamics, a hard linear pressure-overclosure law, and the SFC `persistent_dynamic_sdf_calculix` contact mode. The comparison is intentionally scoped: it does not claim source-level equivalence to every private CalculiX branch, but it does compare the main physical trajectory and contact outputs that can be exported reliably.

Table 7: Official CalculiX contact-law replay on deformed C3D8 contact states. SFC recomputes the current-surface dynamic-SDF gap and pressure from the CalculiX final displacement field.

Case	Law	Max gap error	Max pressure error	Energy claim
`contactenergy`	Linear surface-to-surface	1.52×10^{-16}	1.52×10^{-11}	yes, 3.80×10^{-8} rel.
`contact1`	Exponential node-to-surface	2.00×10^{-14}	6.48×10^{-7}	no, spring-area diagnostic only
`contact3`	Linear node-to-surface	4.11×10^{-9}	2.72×10^{-4}	no, spring-area diagnostic only
`contact6`	Stiff linear node-to-surface	7.00×10^{-13}	3.07×10^{-5}	no, spring-area diagnostic only

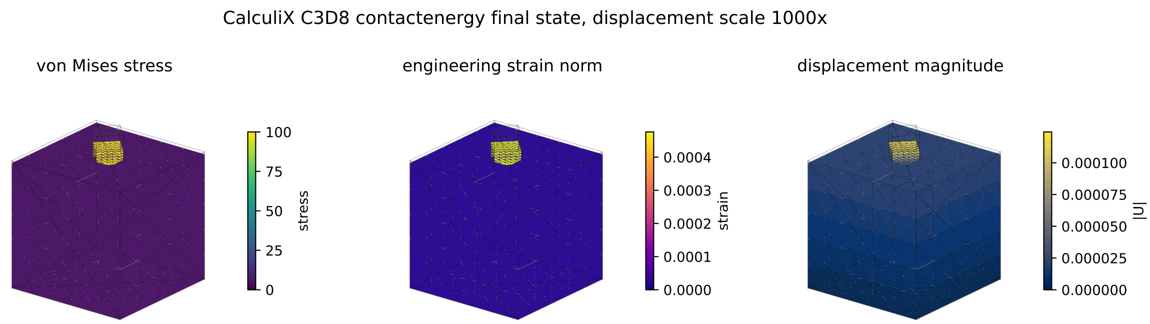

Figure 6: CalculiX `contactenergy.inp` C3D8 final-state visualization. The stress and strain fields are post-processed from the final CalculiX displacement field, averaged to boundary nodes, and interpolated over triangulated C3D8 boundary faces. The deformed shape is plotted with a labeled displacement scale factor to make the small static deformation visible.

This dynamic comparison supports the scoped claim that, with the same external contact law and C3D4 face-to-face contact structure, the dynamic-SDF backend preserves the CalculiX contact trajectory to small errors in the tested block-plane case. The active contact lifecycle is also checked: CalculiX and SFC both reach a maximum CNUM-equivalent value of 14, and the replay-on-CalculiX-displacement CNUM sequence matches the CalculiX sequence in the locked diagnostic output.

5.6 Implementation-level geometry oracle

The external contact experiments above are the main correctness evidence for contact behavior. Separately, we retain a brute-force all-triangle closest-point projection check as an implementation oracle for the SDF query itself. In the final packaged run, the reference used surface resolution 12, with 288 triangles and 144 reference rows. The maximum reported gap error and normal error were both zero for the tested configuration.

This confirms that the spatial-hash candidate pipeline and dynamic surface-distance query reproduce direct closest-point projection for the tested geometry. It does not by itself prove physical contact correctness, and it does not imply global SDF correctness for arbitrary non-manifold or inconsistently oriented geometry.

5.7 Internal contact force-displacement regression

We also retain an internal force-displacement regression against brute-force all-triangle projection on a simple rigid-plane penalty case. Both paths use the same penalty law, so the comparison isolates whether the accelerated query produces the same active set and force for this implementation

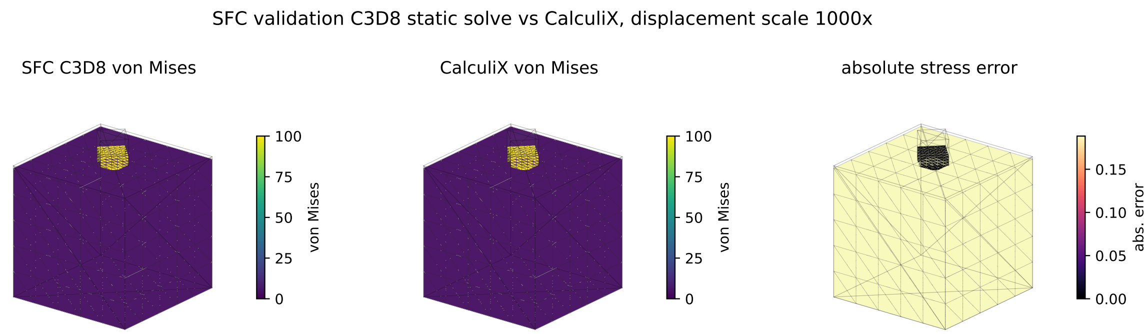

Figure 7: Direct validation-only C3D8 comparison on CalculiX `contactenergy.inp`. The first panel shows the SFC C3D8 static solve, the second shows CalculiX, and the third shows the absolute von Mises stress error. This comparison is included to make the external contact evidence visually auditable; it is not a claim of nonlinear C3D8 dynamic trajectory equivalence.

Table 8: External dynamic C3D8 trajectory replay against CalculiX. CalculiX provides the trajectory; SFC replays the deformed C3D8 current surface with the dynamic SDF query.

Case	Rows	Active rows	Max penetration	RF rel. error	CELS rel. error
Block-plane C3D8	69	22	1.549680×10^{-3}	1.435099×10^{-4}	2.403822×10^{-4}
Block-block C3D8	70	22	1.438760×10^{-3}	2.428073×10^{-4}	2.315966×10^{-4}

test. This table is not used as the main contact validation claim; the external CalculiX comparisons provide that role.

5.8 Material-space SDF baseline

A frozen material-space SDF baseline was evaluated under stretch and shear deformation. This baseline is not the proposed method; it is used only to demonstrate the errors introduced by carrying a reference-space distance function under deformation.

For stretch deformation, the maximum gap error was 5.217391×10^{-3}. For shear deformation, the maximum gap error was 2.708013×10^{-3}, and the maximum normal angle error was 3.580203×10^{-1}. These results support the motivation that a reference/material-space SDF is not sufficient to represent current-configuration distance and normal information under deformation.

5.9 Contact time-history diagnostics

The internal contact diagnostics record time, minimum gap, maximum penetration, active contact count, normal force, contact energy, and action-reaction imbalance. These diagnostics are useful for regression testing and for separating geometry errors from force-assembly errors, but they are not a substitute for the external CalculiX contact-output comparisons above.

5.10 Performance scaling

The final non-quick experiments were run with surface sizes 4, 8, 12, and 16, using five timing repeats per row. The dynamic FEM-induced SDF full pipeline achieved the speedups over brute-force all-triangle projection shown in Table 14.

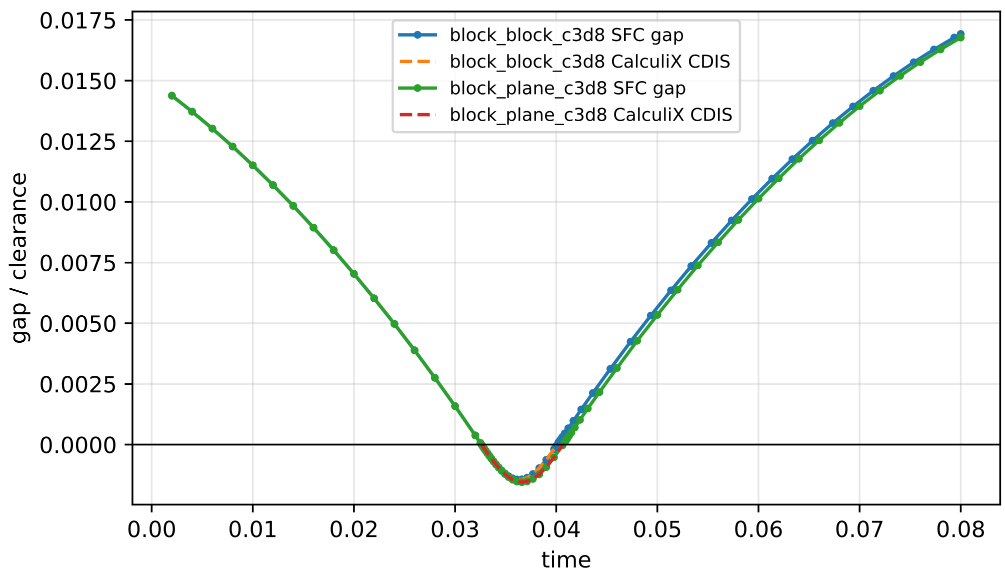

Figure 8: C3D8 dynamic contact trajectory replay: SFC current-surface dynamic-SDF gap compared with CalculiX `CDIS` output where contact rows are available.

Table 9: Native C3D8 dynamic trajectory comparison against CalculiX quick references. Errors are relative L2 trajectory errors except active count.

Case	z_{cm} error	Gap/CDIS error	RF error	CELS error	Active count error
Linear block-plane	8.069672×10^{-4}	1.571721×10^{-2}	2.189529×10^{-17}	0.000000	6
Linear block-block	1.937555×10^{-4}	7.489922×10^{-3}	0.000000	0.000000	4
Geometric nonlinear block-plane	1.115159×10^{-3}	1.956589×10^{-2}	0.000000	0.000000	0
Geometric nonlinear block-block	2.521191×10^{-4}	9.778894×10^{-3}	0.000000	0.000000	0

These results support the claim that, for the tested configurations, the dynamic FEM-induced surface-distance pipeline reduces contact detection cost relative to brute-force all-triangle projection. To check whether this remains meaningful beyond contact-only timing, Phase-7 also measures a small total-step workload that includes FEM assembly, a deterministic sparse linear solve, and contact evaluation. Table 15 shows that the dynamic path remains faster than the brute-force counterpart for the tested sizes.

The comparison is not a claim of superiority over production BVH, IPC, GPU collision detection, or commercial solvers. It shows that, in the current prototype and tested workloads, replacing brute-force all-triangle projection with the spatial-hash dynamic SDF query can reduce both contact-only cost and the measured assembly/solve/contact step cost.

6 Limitations

The current implementation and experiments have several limitations.

First, the production FEM formulation is still narrow. Linear C3D4/TET4 and C3D8 mechanics backends are implemented, and the validation backend now includes clean-room StVK/HHT paths for C3D4 and C3D8. C3D10, corotational formulations, and hyperelasticity are not implemented.

Second, contact is limited to frictionless normal penalty contact. Friction, self-contact, augmented Lagrangian methods, active-set methods, barrier contact, and IPC-style formulations are

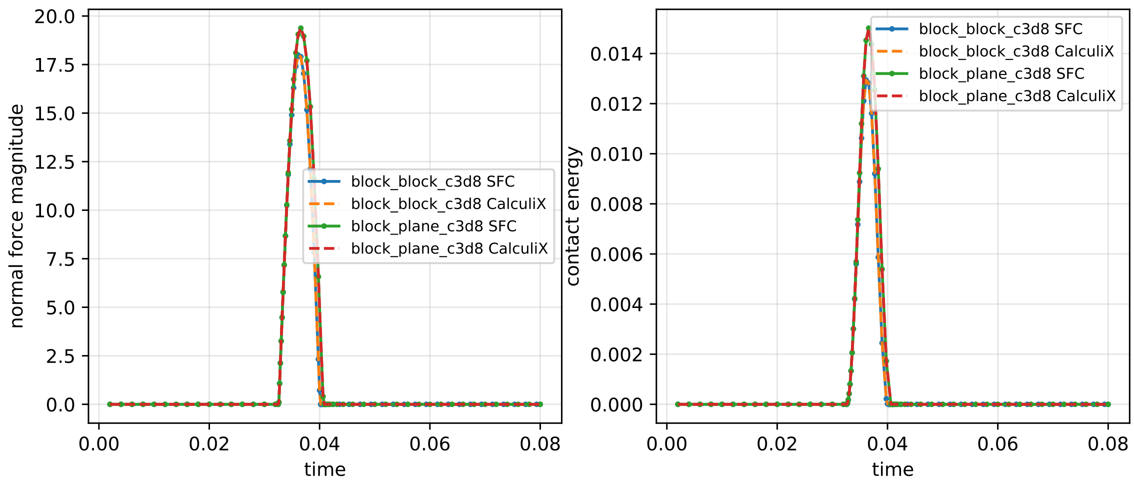

Figure 9: C3D8 dynamic contact trajectory replay: fixed-support reaction force and contact spring energy from CalculiX compared with the SFC dynamic-SDF replay force and penalty energy.

Table 10: Per-case SFC/CalculiX wall-time comparison from the locked Phase-9 quick evidence. Efficiency claims are allowed only when the same row also passes the correctness gate and speedup is greater than one.

Case	SFC time (s)	CalculiX time (s)	Speedup	Efficiency claim
Linear block-plane	1.613157×10^{-1}	9.104603×10^{-1}	5.64×	yes
Linear block-block	3.109808×10^{-1}	1.100008	3.54×	yes
Geometric nonlinear block-plane	4.073683	1.791621	0.44×	no
Geometric nonlinear block-block	9.378789	1.072140	0.11×	no

outside the current scope.

Third, the distance query is an oriented local current-surface distance. The method is validated for outward-oriented closed TET4 surfaces, but it is not a robust global signed-distance method for arbitrary open, non-manifold, or inconsistently oriented geometries.

Fourth, the broad phase is a uniform spatial hash. The reported speedups are measured against brute-force all-triangle projection, not against optimized production BVH implementations.

Finally, timing results are hardware- and implementation-dependent. The final experiment metadata must accompany any performance table derived from these results. The reported numbers should be interpreted as evidence for the current standalone prototype and not as a universal performance bound.

The external evidence is also bounded. The stress/strain check is an analytic affine patch test, and the external open-source FEM comparison uses scikit-fem on matching linear TET4 cantilever models. The contact evidence uses CalculiX output for a static C3D8 contact-energy replay, scoped native C3D8 block-plane/block-block dynamic trajectory comparisons, scoped C3D8 trajectory replay, a scoped dynamic C3D4 block-plane trajectory, and a warped/non-planar C3D8 diagnostic whose metrics are not yet claim-supported. These cases support the current-surface contact query and force/energy replay in the tested settings, but they do not establish frictional contact, self-contact, sphere-drop impact, production BVH, arbitrary curved-surface accuracy, or source-level CalculiX equivalence.

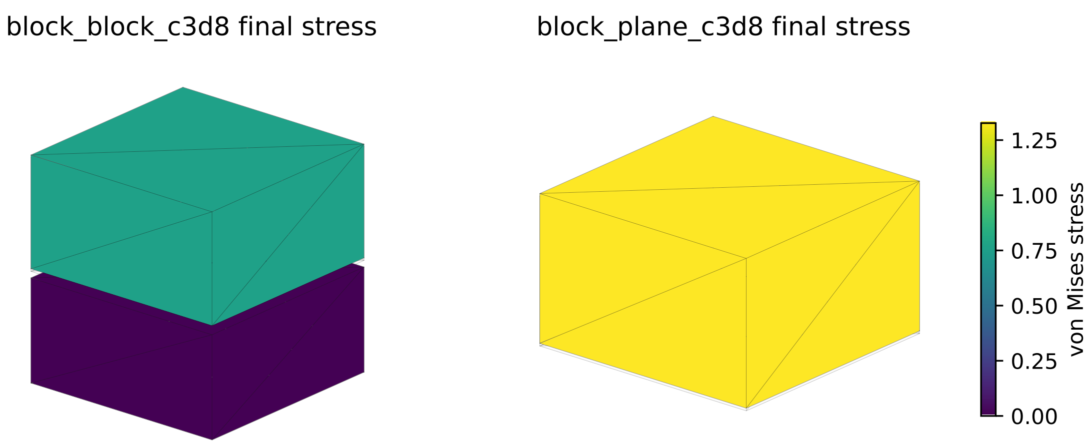

Figure 10: Final C3D8 stress clouds for the generated block-plane and block-block trajectory replay cases. The stress field is displayed on the deformed C3D8 boundary after nodal averaging for visual audit.

Table 11: Dynamic external contact trajectory comparison against CalculiX for the locked 1 s block-plane case.

Metric	CalculiX	SFC dynamic-SDF	Relative/absolute error
Contact activation time	2.200000×10^{-2}	2.200000×10^{-2}	1.39×10^{-17} abs.
Maximum penetration	1.044401×10^{-2}	1.068598×10^{-2}	2.316794×10^{-2}
Peak normal force	1.071145×10^{2}	1.081660×10^{2}	9.816751×10^{-3}
Peak contact energy	4.391052×10^{-1}	4.479852×10^{-1}	2.022301×10^{-2}
Maximum CNUM equivalent	14	14	0.000000 abs.
z_{cm} trajectory	–	–	2.601493×10^{-4} L2 rel.
Contact-zone max von Mises	4.570907×10^{1}	4.588905×10^{1}	3.937603×10^{-3}

7 Conclusion

This paper presented a training-free dynamic surface-distance framework for finite-element contact of deformable bodies. Unlike static SDFs or data-driven dynamic SDF reconstruction, the proposed method induces a local oriented distance directly from the current finite-element boundary. The resulting contact gap, normal, Jacobian, and penalty force are assembled consistently from the current FEM geometry.

A standalone implementation was developed without dependence on Abaqus-generated matrices or data files. The prototype supports registered C3D4/TET4 and C3D8 element backends, spatial-hash contact candidates, dynamic surface-distance queries, local closest-point projection, and frictionless penalty contact. Validation includes analytic linear-elastic stress/strain references, external scikit-fem displacement/stress comparisons, CalculiX static contact-energy replay on a C3D8 surface, native C3D8 block-plane/block-block dynamic trajectory comparisons, CalculiX dynamic C3D8 trajectory replay on block-plane and block-block cases, a CalculiX dynamic C3D4 block-plane trajectory comparison, material-space SDF baseline comparisons, implementation-level closest-point checks, and repeated performance timing.

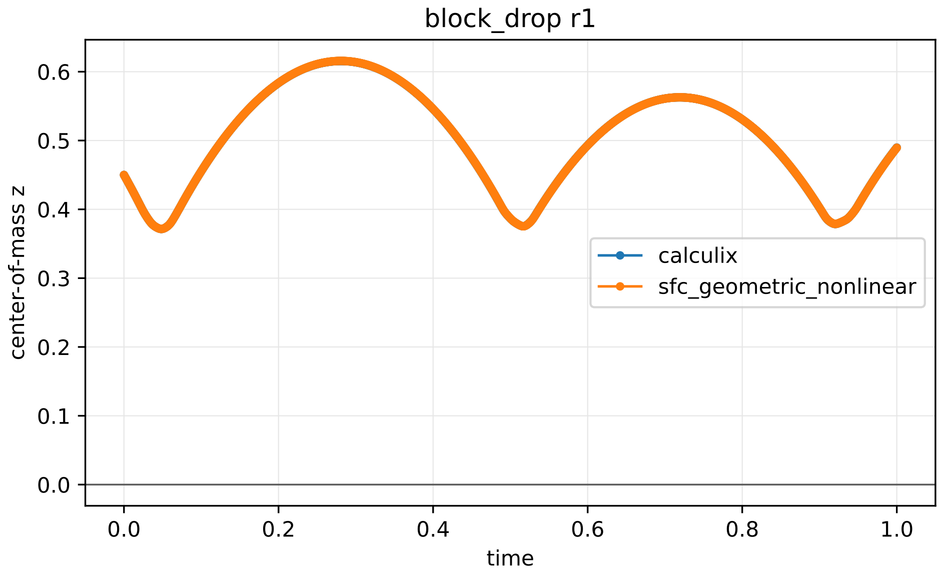

Figure 11: Dynamic block-plane center-of-mass trajectory compared against CalculiX for the locked 1 s contact case.

Table 12: Rigid-plane contact force-displacement comparison. The dynamic SDF path and brute-force CPP reference use the same normal penalty law.

Penetration	Active count	Brute-force force	Dynamic SDF force	Relative error
0.000000×10^{0}	0/0	0.000000×10^{0}	0.000000×10^{0}	0.000000×10^{0}
2.500000×10^{-3}	64/64	8.000000×10^{2}	8.000000×10^{2}	0.000000×10^{0}
5.000000×10^{-3}	64/64	1.600000×10^{3}	1.600000×10^{3}	0.000000×10^{0}
1.000000×10^{-2}	64/64	3.200000×10^{3}	3.200000×10^{3}	0.000000×10^{0}
2.000000×10^{-2}	64/64	6.400000×10^{3}	6.400000×10^{3}	0.000000×10^{0}

In the tested final non-quick experiments, the dynamic FEM-induced surface-distance pipeline achieved 3.58× to 57.07× speedup over brute-force all-triangle projection. These results suggest that current-surface-induced dynamic distance queries can provide an efficient and training-free geometric front end for deformable FEM contact. Future work will extend the framework to C3D10 mechanics, frictional and self-contact, robust global sign handling, production BVH/GPU acceleration, and barrier or augmented-Lagrangian contact formulations.

Evidence Package

The final experiment package is stored under `results/paper_final`. Figures are drawn from `results/paper_final/figures`, tables from `results/paper_final/tables`, and runtime metadata from `results/paper_final/metadata`. The added physical-comparison outputs are stored under `results/phase7`. The locked external evidence used in this draft is stored under `paper/numerical_experiments/scikit_fem_cantilever_external`, `paper/numerical_experiments/calculix_contactenergy_c3d8_replay`, `paper/numerical_experiments/c3d8_contact_trajectory_validation`, `paper/numerical_experiments/phase9_full_contact_validation`,

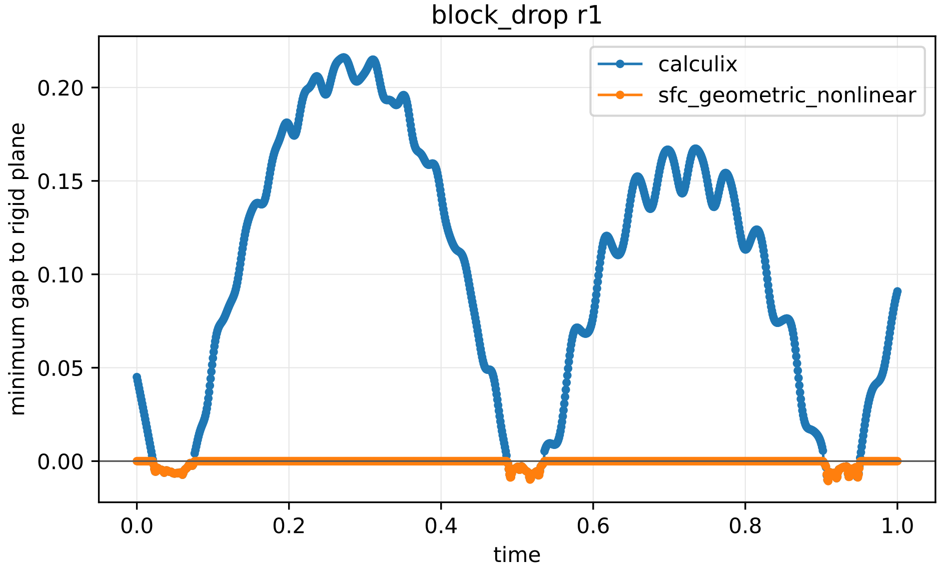

Figure 12: Minimum contact gap history in the dynamic CalculiX comparison. The plot reports the penetration behavior produced by the dynamic-SDF contact backend and the CalculiX reference output.

Table 13: Material-space SDF baseline error under deformation. The baseline is a frozen material-space SDF and is used only for comparison.

Deformation	Rows	Max gap error	Max normal angle error
Stretch	20	5.217391×10^{-3}	0.000000×10^{0}
Shear	20	2.708013×10^{-3}	3.580203×10^{-1}

and `paper/numerical_experiments/block_drop_dynamic_sdf_calculix_1s`. The source handoffs are documented in `docs/phase6_final_experiment_handoff.md`, `docs/phase7_physical_validation_handoff.md`, `docs/c3d8_contact_trajectory_validation.md`, and the corresponding numerical-experiment manifests.

Citation TODO

All placeholder bibliography entries in `references.bib` must be replaced before submission. Required citation groups include FEM node-to-surface contact, mortar contact, penalty contact, SDF contact, deformable or dynamic SDFs, neural/reduced SDFs, broad-phase collision detection, BVHs, and IPC/barrier contact.

References

[1] Todo: replace with broad-phase collision detection references. Placeholder citation for uniform grids, spatial hashing, and BVHs.

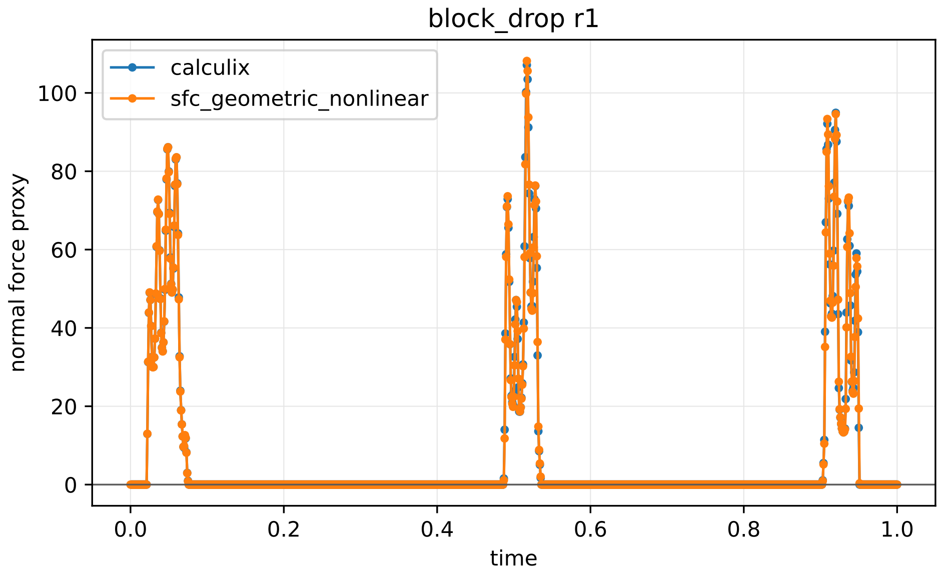

Figure 13: Normal reaction force history for the locked dynamic block-plane comparison. Force agreement is evaluated using CalculiX reaction/contact output where available.

Table 14: Final non-quick performance scaling. Speedup is measured against brute-force all-triangle projection, not against production BVH, IPC, GPU collision detection, or commercial solvers.

Surface res.	Triangles	Query points	Dynamic mean (s)	Speedup
4	32	16	1.13×10^{-2}	3.58×
8	128	64	4.15×10^{-2}	14.99×
12	288	144	9.40×10^{-2}	32.48×
16	512	256	1.69×10^{-1}	57.07×

[2] Todo: replace with deformable or dynamic sdf references. Placeholder citation for dynamic/deformable signed distance fields.

[3] Todo: replace with finite-element contact references. Placeholder citation for FEM contact and node-to-surface contact.

[4] Todo: replace with ipc or barrier-contact references. Placeholder citation for IPC, barrier contact, and related out-of-scope methods.

[5] Todo: replace with mortar contact references. Placeholder citation for mortar contact and surface-to-surface contact.

[6] Todo: replace with neural or reduced sdf references. Placeholder citation for neural SDFs, reduced bases, and data-driven distance fields.

[7] Todo: replace with signed-distance-field contact references. Placeholder citation for signed distance fields in collision and contact.

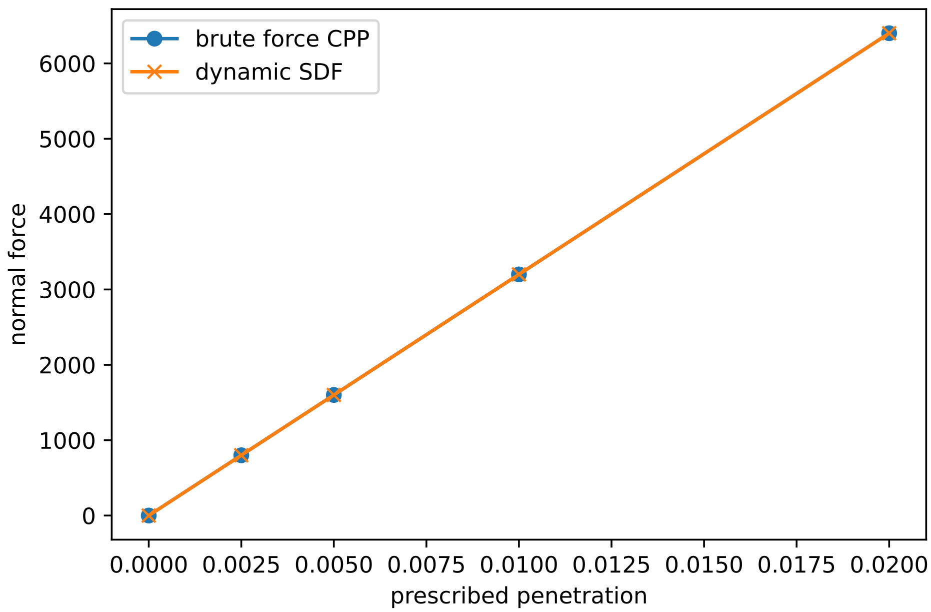

Figure 14: Contact force-displacement comparison against brute-force closest-point projection contact. The curves overlap for the tested rigid-plane normal penalty case.

Table 15: Phase-7 acceleration feasibility. Total-step timing includes FEM assembly, sparse linear solve, and contact evaluation.

Surface res.	Triangles	Contact-only speedup	Total-step speedup
4	32	4.50×	3.24×
8	128	15.55×	13.25×
12	288	33.72×	30.63×

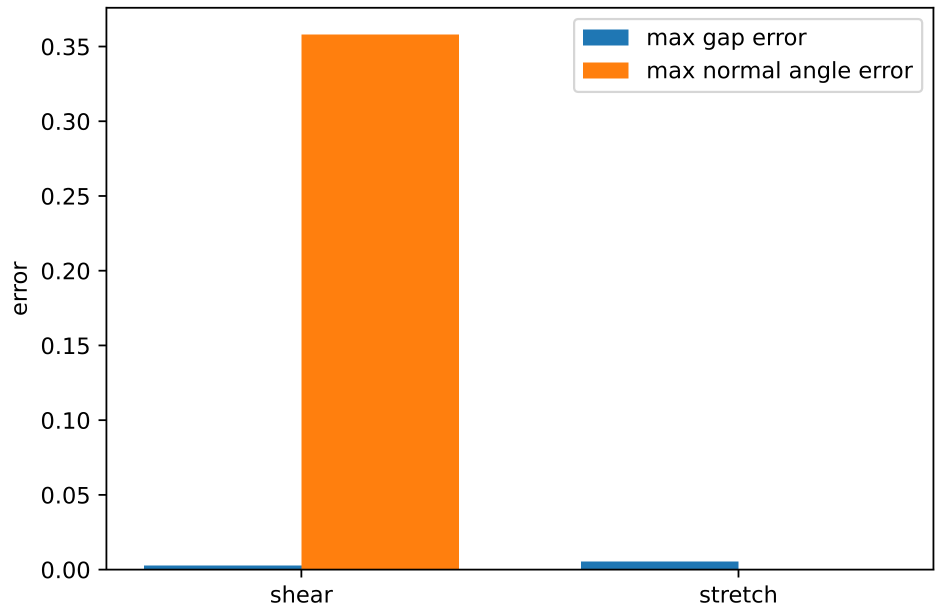

Figure 15: Material-space SDF baseline error under stretch and shear. The proposed method queries the current FEM surface; the baseline carries a frozen material-space distance.

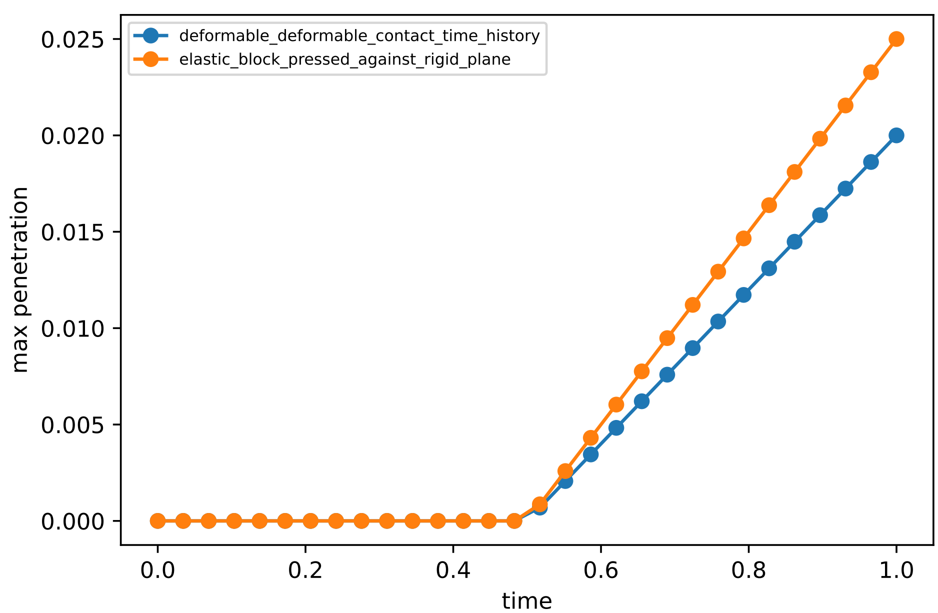

Figure 16: Contact time-history diagnostics from the Phase-6 package. The plotted quantities summarize penetration behavior for the rigid-plane and deformable-deformable contact checks.

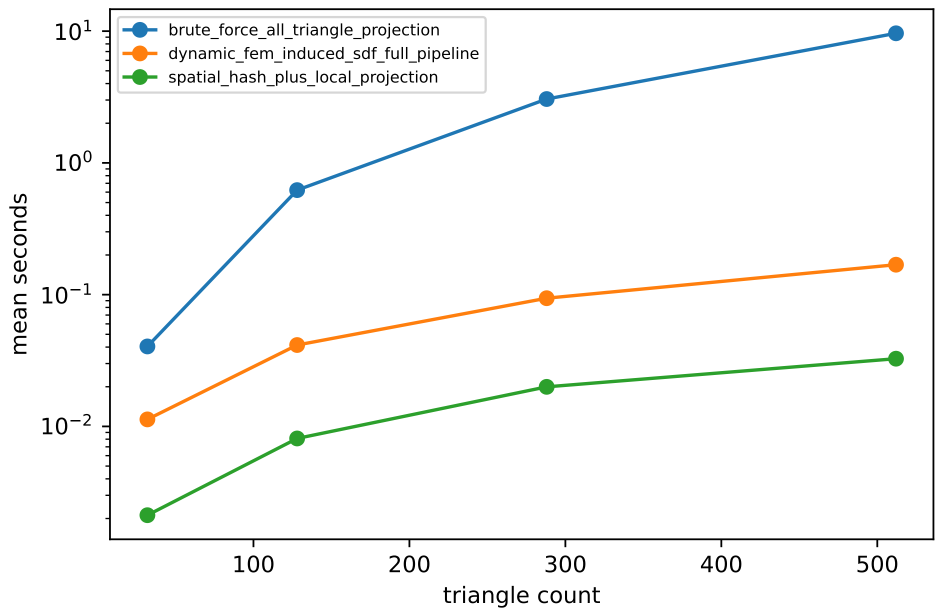

Figure 17: Performance scaling for brute-force all-triangle projection, spatial-hash plus local projection, and the dynamic FEM-induced SDF full pipeline.

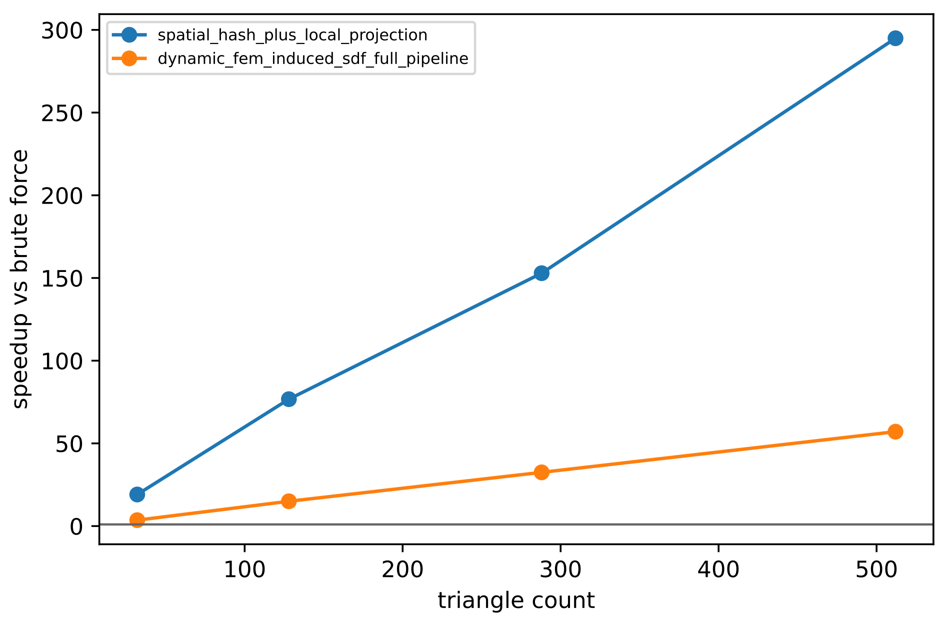

Figure 18: Measured speedup against brute-force all-triangle projection for the tested surface resolutions.

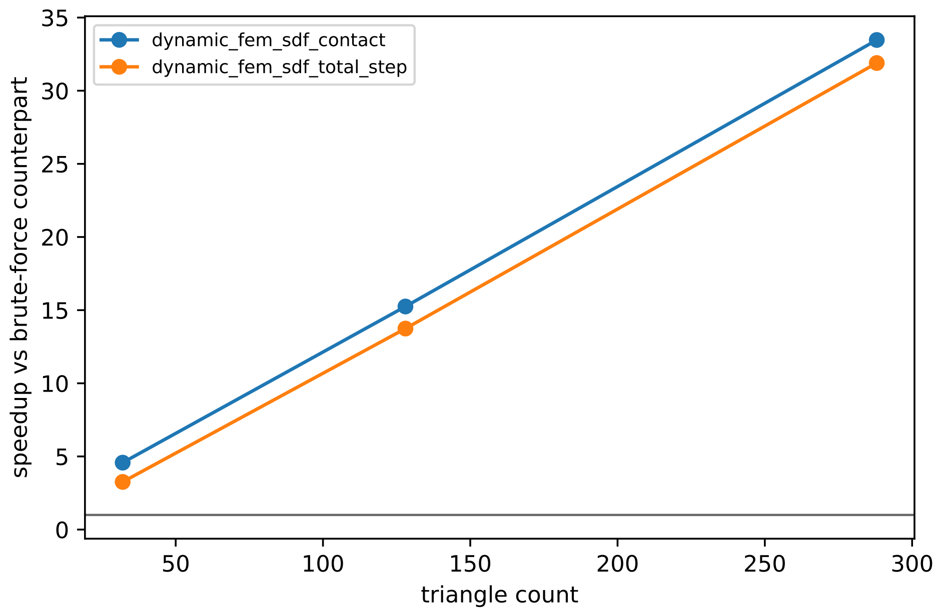

Figure 19: Phase-7 acceleration feasibility. The plot separates contact-only speedup from total-step speedup that includes FEM assembly and a sparse linear solve.